Effectiveness of devices purported to reduce flatus odor.

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OBJECTIVE: A variety of charcoal-containing devices are purported to minimize problems with odoriferous rectal gas; however, the evidence supporting the efficacy of these products is virtually all anecdotal. We objectively evaluated the ability of these devices to adsorb two malodorous, sulfide gases (hydrogen sulfide and methylmercaptan) instilled at the anus. METHODS: Via a tube, 100 ml of nitrogen containing 40 ppm of sulfide gases and 0.5% H(2) was instilled at the anus of six healthy volunteers who wore gas impermeable Mylar pantaloons over their garments. Since H(2) is not adsorbed by charcoal, the fraction of the sulfide gases removed could be determined from the concentration ratio of sulfide gas: H(2) in the pantaloon space relative to the ratio in instilled gas. RESULTS: Measurements with no device in place showed that subjects' garments removed 22.0 +/- 5.3% of the sulfide gases, and results obtained with each device were corrected for this removal. The only product that adsorbed virtually all of the sulfide gases was briefs constructed from an activated carbon fiber fabric. Pads worn inside the underwear removed 55-77% of the sulfide gases. Most cushions were relatively ineffective, adsorbing about 20% of the gases. CONCLUSIONS: The ability of charcoal-containing devices to adsorb odoriferous rectal gases is limited by incomplete exposure of the activated carbon to the gases. Briefs made from carbon fiber are highly effective; pads are less effective, removing 55-77% of the odor; cushions are relatively ineffective.